MIDTERM ASSIGNMENT

Midterm: write a pickup-summary function

Start here: open, write, check

1 · Open

Open Week_04.ipynb.
Go to “Midterm begins” in Session 2.

2 · Write

Complete summarize_request.
Use the supplied fields.
Keep the required function name.

3 · Check

Run the four cases below.
Compare all three result fields.
Then create your ZIP.

Your job is to write one function, not a complete reservation application.

What your function must calculate

Input

A list of batch lines
and reviewed = True or False.
Each line has a count and unit.

Your work

Use a loop to add the counts
and count × unit for each line.
Use branches to choose the label.
Return the three results.

Return

One dictionary:
total_containers
planned_portions
coordination_label

Use valid supplied inputs. CSV parsing and database work are outside this task.

Example: given this input, return this result

```
lines = [{"batch_id": "DEMO",
          "containers": 2,
          "portions_per_container": 3}]
reviewed = True

expected = {
    "total_containers": 2,
    "planned_portions": 6,
    "coordination_label": "standard",
}
```

1. Each list item is one batch line.
2. Multiply containers by the unit on that line.
3. Return values; printing alone is insufficient.

Complete summarize_request(lines, reviewed) with these exact returned field names.

Choose the label using these three rules

Input condition	Required label	Reason
reviewed is False	hold	Human review is missing
Reviewed; portions < 20	standard	Below the workload threshold
Reviewed; portions >= 20	team	Equality is included

Midterm threshold: 20. The separate Week 4 practice variant uses 16.

Example: add both lines before choosing the label

Step	Containers so far	Portions so far
Start	0	0
Line X: 1 × 4	1	4
Line Y: 2 × 6	3	16
Reviewed = True	3	16 → standard

The new line contributes to the running total; it does not replace the earlier total.

Run these four checks on your function

Case and review	Lines: containers × unit	Expected: count / portions / label
A · False	4 × 5	4 / 20 / hold
B · True	3 × 3	3 / 9 / standard
C · True	4 × 5	4 / 20 / team
D · True	2 × 3 and 4 × 4	6 / 22 / team

Write each expected result first; assert all three returned fields in every case.

Do the work in this order

1

1 · Predict

Write the expected totals and label for each case.

2

2 · Implement

Complete your function. Run one case, then the other three.

3

3 · Test

Assert all three fields for every case. Fix any difference.

4

4 · Package

Save the files, write the README and test the extracted ZIP.

If a test fails, locate the first wrong field before changing the code.

Create these three files

File	Put this in it
midterm.py	Your completed summarize_request function.
test_midterm.py	The four inputs, expected results and assertions.
README.md	How to run/tests, expected result, limitation and AI disclosure.
One ZIP	The three files above; test after extracting into a fresh folder.

Submit to submit.progmis@gmail.com. Exact ZIP name to be announced.

README: make the run reproducible

Start

Purpose and fictional-data limit
Python version
Required files and working folder

Run and check

Exact run and test commands
Expected result for one case
How to recognize a passing test

Explain

Loop and review-first rule
One limitation
AI use or non-use

A third party should be able to run the work without asking you for missing steps.

AI is optional; understanding is required

No AI used

State: “I did not use AI for this assignment.”

If AI helped

Name the tool and stages.
Show one help request.
Explain one suggestion you accepted, changed or rejected.

In either case

Run your own tests.
Trace your own function.
Explain a choice or small change.

Marks reward accurate disclosure and demonstrated understanding.

You are ready when you can tick every row

Check	What you can show
Function	All three returned fields have the required names.
Four cases	A, B, C and D all match every expected field.
Fresh folder	The commands in README work after ZIP extraction.
Your explanation	You can trace a line, a total and the review-first decision.
Disclosure	README states no AI use, or describes the help and your judgment.

Keep the ZIP, notebook and a backup. Be ready for your announced explanation slot.

Proposed rubric: program and tests

Criterion	Points	Evidence
Totals and units	6	Correct loop and both totals
Review-first rule	6	Correct hold, standard and team
Common tests	6	Four cases; all fields asserted
Structure and names	4	Clear function and returned values

Program and test evidence: 22 of the proposed 30 points.

Proposed rubric: reproducibility and explanation

Criterion	Points	Evidence
README and ZIP	4	Fresh extraction runs from the guide
Individual explanation	3	Run, trace and explain a choice
AI-use disclosure	1	Honest statement and judgment
Total assessment	30	22 program/test + 8 here

Optional bonus details remain undecided; complete the standard task first.

Half-week timing and individual explanation

90 minutes: implement

Session 2: last 45 minutes
Session 3: first 45 minutes
Work on the required function.

45 minutes: finish

30 minutes: tests, README, ZIP
15 minutes: review and hand-in
Keep a separate backup.

Individual explanation

Five minutes per student.
Slots announced for the cohort
and available examiners.

Week 4 protects 135 minutes for the Midterm. Calendar dates to be announced. Submit to submit.progmis@gmail.com.